THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ZONAL MOCK EXAMINATION

FORM FOUR-MARKING GUIDE

PHYSICS 2A

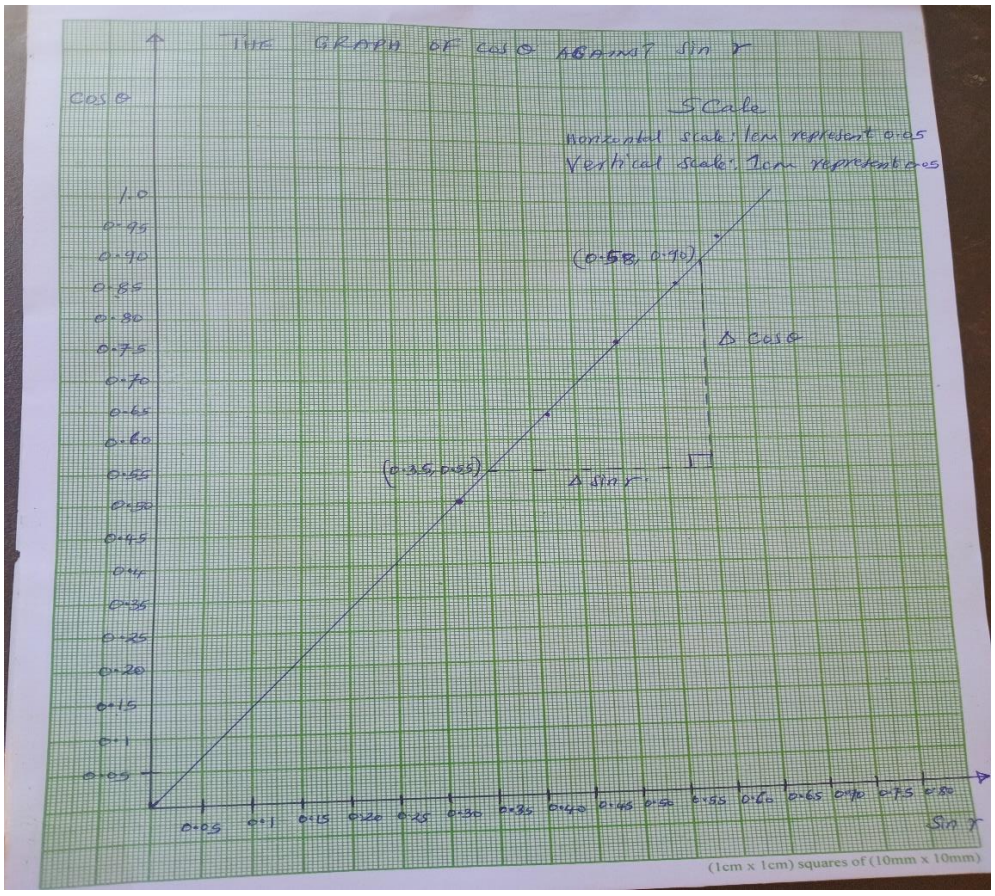
Solution

TABLE OF RESULT

b.

$\theta°$	r°	Cos $\theta°$	Sin r°
20	38	0.9397	0.6157
30	35	0.8660	0.5735
40	30	0.7660	0.5000
50	25	0.6428	0.4226
60	19	0.5000	0.3256

(0.5@15=7.5 Marks)



Title 01 Mark

Scale 01 Mark

SI=01 Mark

Bl = 01 Mark

T.p $\frac{1}{2}$ @ 05=$2\frac{1}{2}$ Marks

d. Slope (n) = $\frac{\Delta Cos\theta}{\Delta Sin\, r}$ $\frac{1}{2}$ ***Mark***

= $\frac{0.90-0.55}{0.58-0.35}$ $\frac{1}{2}$ ***Mark***

= 1.5217

The slope of the graph is 1.52. ***02 Mark.***

e. The physical meaning of the slope of the graph is the refractive index of the glass block.

(***01 Mark)***

f. from the graph

$\frac{i}{n}$=Sin C

Sin c = $\frac{1}{1.52}$ ***01 Mark***

C=$Sin^{-1}\left[\frac{1}{1.52}\right]$ **01 Mark**

C=sin^{-1} (0.65789)

C=41.1395

C=41.14° **01 Mark**

Sources of error are

i. Parallax error in measuring the angle by using protractor.
ii. Parallax error in replacing the glass block
iii. Parallax error in locating a pin.

<u>01@02 Marks (Any two)</u>

Precautions to be taken

i. To be carefully in measuring the angle by protractor.
ii. To be carefully in replacing the glass block
iii. To be carefully in locating pins.

01@02 Marks (Any two)

2.

TABLE OF RESULTS

R((Ω)	d(cm)	$\frac{100}{d}$
1	66.6	1.5
2	50	2
3	40	2.5
4	33.3	3
5	29	3.4

$\frac{1}{2}$ ***@10=05 Marks.***

THEORY

R∝L

R∝ KL

$\frac{R}{L}$ =K

$\frac{R1}{L1} = \frac{R2}{L2}$ ***01 Mark.***

Let L1 = d

R1 = x

R2 = R

L2 = 100-d

$\frac{x}{d} = \frac{R}{100-d}$ ***01 Mark.***

dR=(100-d)x

$\frac{dR}{d} = \frac{100x - dx}{d \quad d}$

R= x $\frac{100-x}{d}$ **01 Mark.**

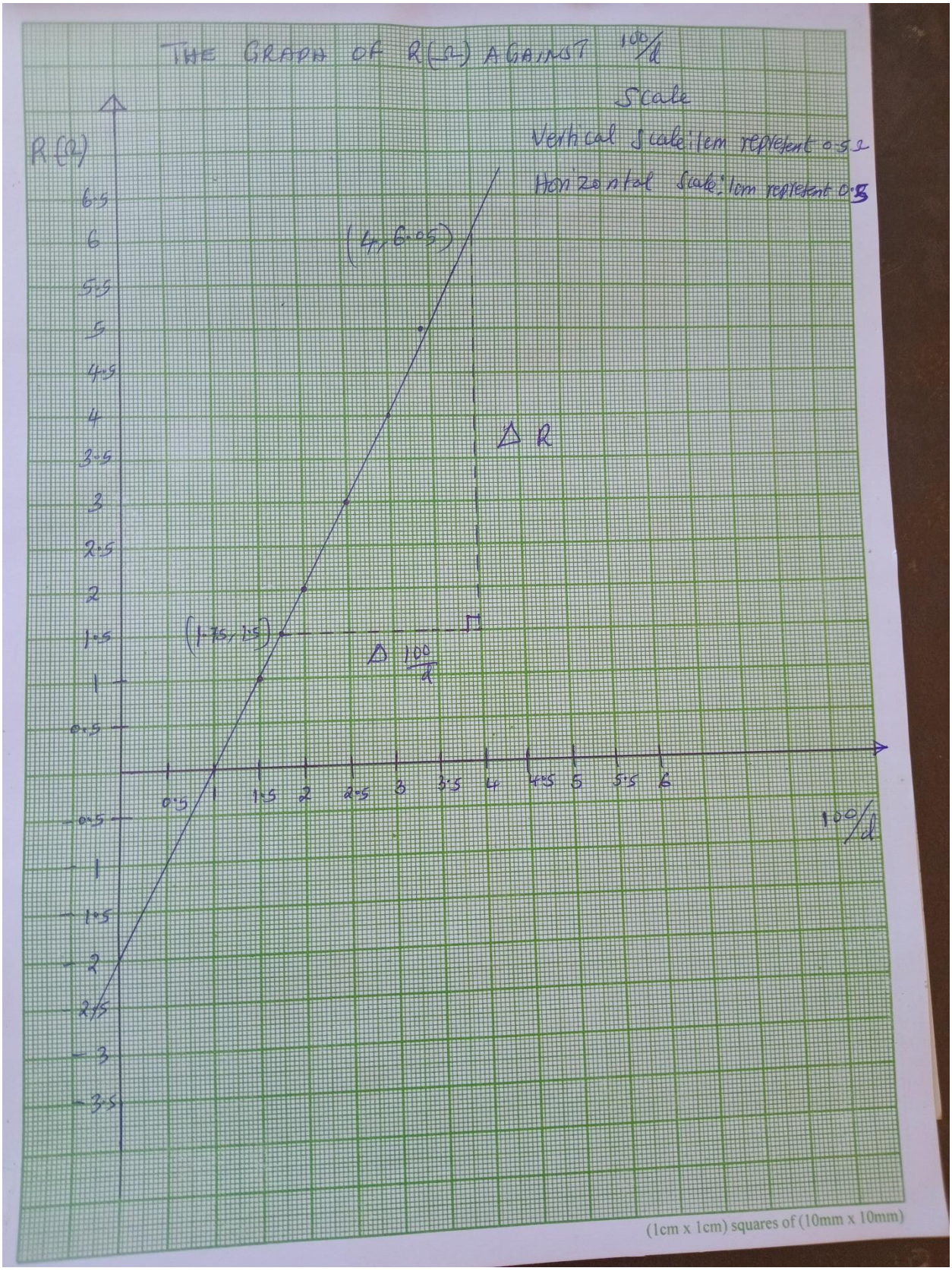
Y= mx – c

M(slope) = x (UNKNOWN Resistance) **0.5 Mark.**

Also

$^{-}C = X$

C=X (Unkown resistance) **0.5 Mark.**



Title 01 Mark

Scale 01 Mark

SI 01Mark.

Bl 01 Mark.

Tp 2.5 Marks.

(b) from the graph

Slope (M) = $\frac{\Delta R}{\Delta^{100}/_{d}}$ **01Mark**

$=\frac{(6.05-1.5)}{(4-1.75)}$ **01 Mark**

$=\frac{4.55}{2.25}$

Slope (M=2.02 Ω

The slope of the graph is 2.02 Ω **02 Marks.**

(c) From the graph

Ro on R-axis is 2Ω **02 Marks.**

(d) To deduce value of x

Solution

Slope of the graph (M) = x

Slope = 2.02Ω

X=R-intercept (Ro)=2

$\frac{2+2.02}{2}$ **0.5 Mark.**

X=2.01$\approx$ 2Ω

The value of x is 2Ω **01 Mark.**

(e) The suitable title of this experiment is determination of unknown resistance by using metre bridge. **02 Marks.**